ECE 6950 PSET 7

Q1

a) Recall Eq 6.3-6.6: (note: Ω real, so $\Omega^* = \Omega$)

$$\dot{\rho}_{ee} = -\Gamma \rho_{ee} - i\Omega^* \rho_{eg} + i\Omega \rho_{ge} \qquad \dot{\rho}_{gg} = \Gamma \rho_{ee} + i\Omega^* \rho_{eg} - i\Omega \rho_{ge}$$

$$\dot{\rho}_{ge} = -\frac{\Gamma}{2}\rho_{ge} - i\delta\rho_{ge} + i\Omega^*(\rho_{ee} - \rho_{gg}) \qquad \dot{\rho}_{eg} = -\frac{\Gamma}{2}\rho_{eg} + i\delta\rho_{eg} - i\Omega(\rho_{ee} - \rho_{gg})$$

$$\dot{\rho}_{ee} = -2\Gamma\rho_{ee} + i\frac{\Omega}{2}\rho_{eg} - i\frac{\Omega}{2}\rho_{ge} \qquad \dot{\rho}_{ge} = -\Gamma\rho_{ge} - i\frac{\Omega}{2}(\rho_{ee} - \rho_{gg})$$

$$\Rightarrow \rho_{ee} + \rho_{gg} = 1 \Rightarrow \dot{\rho}_{gg} = -\dot{\rho}_{ee}, \quad \dot{\rho}_{ge} = \dot{\rho}_{eg}^*$$

Note: $\langle \sigma_z \rangle = \langle (|e\rangle\langle e| - |g\rangle\langle g|) \rangle = \rho_{ee} - \rho_{gg} = w$

$$\langle \sigma_y \rangle = i\langle (\sigma_- - \sigma_+) \rangle = i(\rho_{ge} - \rho_{eg}) = v$$

$$\langle \sigma_x \rangle = \langle \sigma_+ + \sigma_- \rangle = \rho_{eg} + \rho_{ge} = u$$

Find $\dot{w} = \dot{\rho}_{ee} - \dot{\rho}_{gg} = -4\Gamma\rho_{ee} + i\Omega\rho_{eg} - i\Omega\rho_{ge}$

$$= -2\Gamma\rho_{ee} - 2\Gamma(1 - \rho_{gg}) + i\Omega\rho_{eg} - i\Omega\rho_{ge}$$

$$\dot{w} = -2\Gamma - 2\Gamma w + \Omega v = \boxed{\dot{w} = 2v - 2\Gamma(w+1)}$$ As desired

$$\dot{v} = i(\dot{\rho}_{ge} - \dot{\rho}_{eg}) = i\left(-\Gamma\rho_{ge} - i\frac{\Omega}{2}\underbrace{(\rho_{ee} - \rho_{gg})}_{w} + \Gamma\rho_{eg} - i\frac{\Omega}{2}w\right)$$

$$\boxed{\dot{v} = -\Gamma v - \Omega w}$$ As desired

$$\dot{u} = \dot{\rho}_{eg} + \dot{\rho}_{ge} = \dot{\rho}_{eg}^* + \dot{\rho}_{eg} = -\Gamma\rho_{ge} - \cancel{i\frac{\Omega}{2}(\rho_{ee} - \rho_{gg})} - \Gamma\rho_{eg} + \cancel{i\frac{\Omega}{2}(\rho_{ee} - \rho_{gg})}$$

$$= -\Gamma(\rho_{ge} - \rho_{eg}) = -\Gamma u$$

$$\boxed{\dot{u} = -\Gamma u}$$ As desired.

Collecting,

$$\boxed{\begin{aligned} \dot{w} &= 2v - 2\Gamma(w+1) \\ \dot{u} &= -\Gamma u \\ \dot{v} &= -\Gamma v - \Omega w \end{aligned}}$$

b) From 7.61, $\langle \hat{O}(t+\tau)\rangle = \sum_j a_j(\tau)\langle O_j(t)\rangle$

$$\Rightarrow \langle \hat{P}\,\hat{O}(t+\tau)\hat{Q}\rangle = \sum_j a_j(\tau)\langle P(t)O_j(t)\,Q(t)\rangle \quad \text{by linearity}$$

So, $\langle \sigma_x(t+\tau)\,\sigma_-(t)\rangle = \sum_j a_j(\tau)\langle \sigma_j(t)\,\sigma_-(t)\rangle$

Note: $\sigma_-\sigma_- = 0 = \sigma_+\sigma_+ \qquad \sigma_-\sigma_+ = |g\rangle\langle g| \qquad \sigma_+\sigma_- = |e\rangle\langle e|$

$$\langle \sigma_x(t+\tau)\sigma_-(t)\rangle = u(\tau)\langle |e\rangle\langle e|\rangle - \cancel{v(\tau)\langle |e\rangle\langle e|\rangle} - \cancel{w(\tau)\langle |g\rangle\langle e|\rangle}$$

$$\langle \sigma_y(t+\tau)\sigma_-(t)\rangle = \cancel{u(\tau)\langle |e\rangle\langle e|\rangle} - v(\tau)\langle |e\rangle\langle e|\rangle - \cancel{w(\tau)\langle |g\rangle\langle e|\rangle}$$

$$\langle \sigma_x(t+\tau)\sigma_-(t)\rangle = u(\tau)\langle |e\rangle\langle e|\rangle$$

$$\langle \sigma_y(t+\tau)\sigma_-(t)\rangle = -iv(\tau)\langle |e\rangle\langle e|\rangle$$

c) $\lim_{t\to\infty}\langle \sigma_+(t+\tau)\sigma_-(t)\rangle =$ $\qquad \sigma_+ = (\hat{\sigma}_x + i\hat{\sigma}_y)/2$

$$= \frac{1}{2}\langle \sigma_x(t+\tau)\sigma_-(t)\rangle + \frac{i}{2}\langle \sigma_y(t+\tau)\sigma_-(t)\rangle$$

$$= \frac{1}{2}\sigma_x(\tau)\langle \sigma_x + \sigma_x^*\rangle(t) + \frac{i}{2}\sigma_y(\tau)\langle \sigma_y^* + \sigma_y\rangle(t)$$

From a,

$$\dot{w} = 2v - 2\Gamma(w+1) \qquad (\Omega^2 - 2\Gamma^2)\cos()$$

$$\dot{u} = -\Gamma u$$

$$\dot{v} = -\Gamma v - \Omega w \qquad (\Omega^2 + 2\Gamma^2)\,v(0) - 2\Gamma\Omega\cos$$

$$\Rightarrow \quad w(\infty) = -\frac{2\Gamma^2}{\Omega^2+2\Gamma^2} \qquad \frac{\Omega^2}{2}\cdot\frac{1}{\Omega^2+2\Gamma^2}$$

$$u(\infty) = 0$$

$$v(\infty) = \frac{2\Gamma\Omega}{\Omega^2+2\Gamma^2}$$

c) Making those above substitutions:

$$w(\infty) = -\frac{2\Gamma^2}{\Omega^2+2\Gamma^2}$$

$$u(\infty) = 0 \quad \Rightarrow \quad u(t)\,u(\tau)\ldots$$

$$v(\infty) = \frac{2\Gamma\Omega}{\Omega^2+2\Gamma^2}$$

$$= \frac{\Omega^2}{4}\left[\frac{e^{-\Gamma\tau}}{\Omega^2+2\Gamma^2} + \frac{1}{(\Omega^2+2\Gamma^2)^2}\left(4\Gamma^2 + e^{-3\Gamma\tau/2}\left((-\Omega^2-2\Gamma^2)\cos(\Omega_\Gamma\tau) + \ldots\right.\right.\right.$$

Seems like it matches at a glance...

d) This is pretty brutal,

Rewriting $\sigma_\Gamma \to \sigma$.

We sub this in and shove it into Wolfram. Luckily, most of it is imaginary.

$$\Rightarrow \int_0^\infty \langle \sigma_+(t+\tau)\sigma_-(t)\rangle e^{-i\omega\tau}\, d\tau$$

No normalization

$\Rightarrow$ By Wolfram,

$$F(\ldots) = \frac{\sqrt{\frac{\pi}{2}}\,\Gamma^2\Omega^2\delta(\omega)}{2(2\Gamma^2+\Omega^2)} + \cancel{\frac{\sqrt{\frac{\pi}{2}}\,\Omega^2\,\delta(\omega+i\Gamma)}{4\Gamma^2+2\Omega^2}} \to 0$$

$$+ \ldots (\text{imaginary}) \ldots + \sqrt{\frac{\pi}{2}}\,\Gamma^2\Omega^2 i$$

$$\Rightarrow 2\,\mathrm{Re}(F(\ldots)) = \frac{\sqrt{\frac{\pi}{2}}\,\Gamma^2\Omega^2\delta(\omega)}{2(2\Gamma^2+\Omega^2)}$$

$\Rightarrow$ chain of δ functions.

Does not really match intuition...